THE INDIAN INTERNATIONAL SCHOOL,DSO
Worksheet – Chapter 8: Application of Integrals (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	The area of the region bounded by $y = x^2$, the x-axis and the lines $x = 0$ and $x = 3$ is: (A) 3 sq units (B) 9 sq units (C) 27 sq units (D) 6 sq units	1
2.	The area enclosed by the circle $x^2 + y^2 = 4$ is: (A) 2π sq units (B) 4π sq units (C) 16π sq units (D) π sq units	1
3.	The area enclosed by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is: (A) 6π sq units (B) 36π sq units (C) 13π sq units (D) 12π sq units	1
4.	The area of the region bounded by $y = \cos x$, the x-axis, $x = 0$ and $x = \frac{\pi}{2}$ is: (A) 2 sq units (B) 1 sq unit (C) $\frac{\pi}{2}$ sq units (D) 0	1

Q No.	Question	Marks
5.	The area of the region bounded by the curve $y = x$ and the x-axis, between $x = 0$ and $x = 2$, is: (A) 2 sq units (B) $\frac{1}{2}$ sq unit (C) 1 sq unit (D) 4 sq units	1
6.	**Assertion (A):** The area bounded by $y = \sin x$ and the x-axis between $x = 0$ and $x = 2\pi$ is 4 sq units. **Reason (R):** $\int_0^{2\pi} \sin x \, dx = 0$. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Find the area of the region bounded by $y^2 = x$, the x-axis and the lines $x = 1$ and $x = 4$, lying in the first quadrant.	2
	LEVEL 2 **Q 8 – Q 14**	**18**
8.	Find the area of the region bounded by $y^2 = 9x$, $x = 2$, $x = 4$ and the x-axis in the first quadrant.	3
9.	Find the area of the region bounded by $x^2 = 4y$, $y = 2$, $y = 4$ and the y-axis in the first quadrant.	3
10.	Using integration, find the area of the region bounded by the line $y = 3x + 2$, the x-axis and the ordinates $x = -1$ and $x = 1$.	3
11.	Find the area of the region in the first quadrant bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the coordinate axes.	2
12.	Find the area of the region bounded by the circle $x^2 + y^2 = 16$, the x-axis and the lines $x = 0$ and $x = 2$, lying in the first quadrant.	3
13.	Find the area bounded by $y = \sin x$ and the x-axis between $x = 0$ and $x = \frac{3\pi}{2}$.	2
14.	Find the area of the region bounded by $y = x^3$, the x-axis and the lines $x = -2$ and $x = 1$.	2
	LEVEL 3 **Q 15 – Q 20**	**27**
15.	Find the area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$.	5
16.	Using integration, find the area of the triangle whose vertices are $(1,0)$, $(2,2)$ and $(3,1)$.	5
17.	Find the area of the region bounded by the curve $y = 4x^2$ and the line $y = 8x + 12$.	5
18.	Using integration, find the area of the region bounded by the curve $4x^2 + y^2 = 36$.	5

Q No.	Question	Marks
19.	Find the area of the region bounded by the parabola $y^2 = 4ax$ and its latus rectum.	3
20.	**Case Study:** A running track in a sports complex is in the shape of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ (units in metres). The region inside the track is to be grassed. **(i)** Find the total area enclosed by the track. **[1]** **(ii)** Find the area of the region in the first quadrant enclosed by the track and the axes. **[1]** **(iii)** Find the area of the region inside the track between $x = 0$ and $x = 3$ in the first quadrant. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 8: Application of Integrals

Q No.	Answer / Key Steps
	LEVEL 1
1.	(B) $\int_0^3 x^2\,dx = 9$
2.	(B) $\pi r^2 = 4\pi$
3.	(A) $\pi ab = 6\pi$
4.	(B) $\int_0^{\pi/2} \cos x\,dx = 1$
5.	(A) $\int_0^2 x\ dx = 2$
6.	(B) Both true, but R is not the explanation (area needs $\lvert\sin x\rvert$).
7.	$\int_1^4 \sqrt{x}\ dx = \frac{2}{3}(8-1) = \frac{14}{3}$ sq units
	LEVEL 2
8.	$\int_2^4 3\sqrt{x}\,dx = 16 - 4\sqrt{2}$ sq units
9.	$\int_2^4 2\sqrt{y}\,dy = \frac{32-8\sqrt{2}}{3}$ sq units
10.	Line meets x-axis at $x = -\frac{2}{3}$. Area $= \frac{1}{6} + \frac{25}{6} = \frac{13}{3}$ sq units
11.	$\frac{3}{4}\int_0^4 \sqrt{16-x^2}\ dx = \frac{3}{4}(4\pi) = 3\pi$ sq units
12.	$\left[\frac{x}{2}\sqrt{16-x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^2 = 2\sqrt{3} + \frac{4\pi}{3}$ sq units
13.	$\int_0^{\pi} \sin x\,dx + \left\lvert\int_{\pi}^{3\pi/2} \sin x\,dx\right\rvert = 2 + 1 = 3$ sq units
14.	$\left\lvert\int_{-2}^0 x^3\,dx\right\rvert + \int_0^1 x^3\,dx = 4 + \frac{1}{4} = \frac{17}{4}$ sq units
	LEVEL 3
15.	Intersection $(4,4)$: $\int_0^4 x\ dx + \int_4^{4\sqrt{2}} \sqrt{32-x^2}\ dx = 8 + (4\pi - 8) = 4\pi$ sq units
16.	Lines: $AB: y = 2x-2$, $BC: y = 4-x$, $AC: y = \frac{x-1}{2}$. Area $= \int_1^2 (2x-2)\,dx + \int_2^3 (4-x)\,dx - \int_1^3 \frac{x-1}{2}\,dx = 1 + \frac{3}{2} - 1 = \frac{3}{2}$ sq units
17.	Intersections at $x = -1, 3$. Area $= \int_{-1}^3 (8x + 12 - 4x^2)\,dx = \frac{128}{3}$ sq units
18.	$\frac{x^2}{9} + \frac{y^2}{36} = 1$; area $= 4\int_0^3 2\sqrt{9-x^2}\,dx = 8 \cdot \frac{9\pi}{4} = 18\pi$ sq units
19.	$2\int_0^a 2\sqrt{ax}\,dx = \frac{8a^2}{3}$ sq units
20.	(i) $\pi(6)(4) = 24\pi$ m². (ii) 6π m². (iii) $\frac{2}{3}\int_0^3 \sqrt{36-x^2}\ dx = 3\sqrt{3} + 2\pi$ m²